



Contact

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Address

Bassano del Grappa, VI
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Italy

Skills

Languages

Italian (Mother tongue),
English (Fluent)

Lab

EEG-based MI Decoding,
BCI for Human-Robot
Interaction, Experimental
Design & Data Analysis

Software

Linux, LaTeX, ROS/ROS2

Programming

C++, Matlab, Python3

Job related

Project Management &
Critical Thinking,
Collaborative & Organized,
Creative Problem-Solving,

Personal

Cooking, Hiking, Wrestling

Piero Simonetto

Control System Engineer

About Me:

Ph.D. Candidate in Information Engineering specializing in core Robotics, HRI, and BCI. Expert in autonomous navigation, advanced control, and integrating real-time neural decoding with physical hardware. Proven track record of international collaboration (UT Austin) and deploying robust robotic solutions in dynamic environments.

Experience:

Ph.D. RESEARCHER — NEUROROBOTICS University of Padua

02/01/2023 - present, IASLAB at unipd

- Designed and implemented an end-to-end BCI-driven navigation system for smart wheelchairs, integrating real-time EEG decoding with robotic autonomous navigation.
- Leveraged C++, ROS/ROS2, and advanced control methodologies to fuse user intent with environment perception safely and comfortably.

VISITING RESEARCHER — NEUROROBOTICS UT at Austin, Ph.D. Mobility period

11/01/2024 - 30/04/2025, José del R. Millán lab

- Developed and validated Error-Related Potential detection algorithms for powered wheelchair teleoperation, enhancing safety and user control.

BCI RESEARCHER Cybathlon BCI race - WHI team (2nd Position Worldwide)

27/10/2024, ETH Zurich

- Optimized Motor Imagery online decoding strategies under high-pressure competitive settings, significantly improving control robustness and reducing latency.

WORKSHOP ORGANIZER WS5 at Summer School on Neurorehabilitation

08/06/2025 - 13/06/2025, SSRN2025

- Organized and managed a comprehensive workshop on connecting Brain-Computer Interfaces to real-world physical devices.

TEACHING ASSISTANT University of Padua

01/10/2025 - 28/01/2026 & 02/10/2023 - 02/02/2024, Intelligent Robotics

- Instructed students in Intelligent Robotics laboratory sessions, mentoring them on deploying robots in dynamic environments using ROS/ROS2 and C++.

ERASMUS STUDENT UCLouvain University

09/2021 - 01/2022, Louvain-la-Neuve

- Focused on Bioengineering & Neurosciences research during an international exchange program.

IT INTERN Reflex&Allen

08/2016 - 09/2016, Wales, UK

- Assisted in company IT infrastructure management and hardware/software troubleshooting.

Selected Projects:

SHARED AUTONOMY WHEELCHAIR Hierarchical Control Framework

- Developed a control system to balance user agency with autonomous safety, processing imperfect commands to execute smooth, collision-free trajectories in cluttered environments.
- Implemented a hierarchical control framework integrating NMPC and Multi-point Virtual APF, accounting for the full physical geometry of the wheelchair.

INTENT DECODING Context-Aware Framework for Human-Robot Interaction

- Solved the ambiguity of inherently noisy intention signals by moving away from isolated decoding and introducing structured environmental context.
- Designed a generalized probabilistic framework for multisensory fusion, significantly increasing action-prediction accuracy by filtering inputs based on what actions are meaningful in the robot's surroundings.

PUBLICATIONS Full list available on Google Scholar / Personal Website

Education:

02/2023 - present, University of Padua Ph.D.: Information Engineering

09/2020 - 10/2022, University of Padua LM-25: Control System Engineering

10/2017 - 09/2020, University of Padua L-8: Ingegneria dell'Informazione

09/2012 - 07/2017, I.T.I.S. E. Fermi High school on Computer Science